

Relation Algebra

§6. Recursive Programs

§6.1. Digits of a number

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$.
 $(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

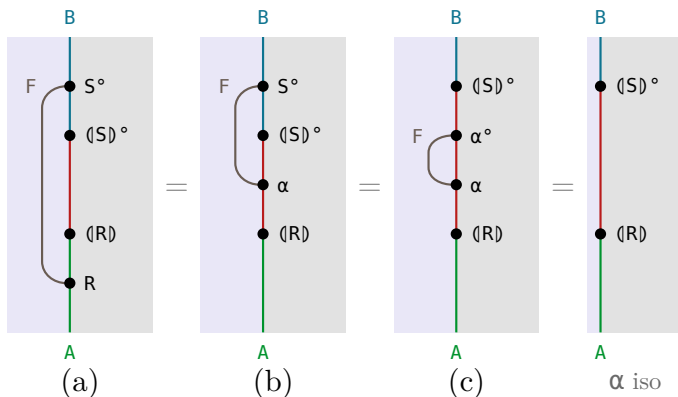
§6.3. Hylomorphisms

$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



(a) the body at $(\mathbb{S})^\circ(\mathbb{R})$

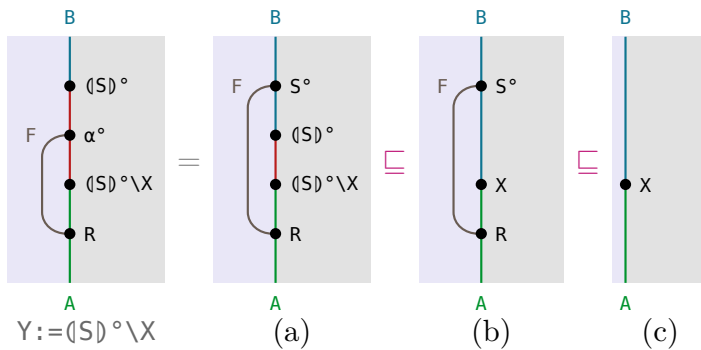
$F(RS) = F(R)F(S) \triangleq$ relator

(b) $F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq$ reduce

(c) $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq$ reduce, relator-laws

$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



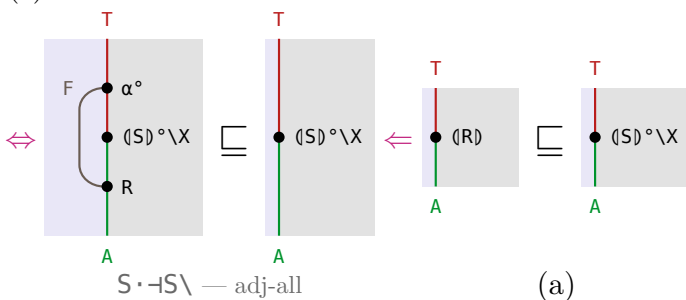
$Y := (\mathbb{S})^\circ \setminus X$

(a) $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$

$F(RS) = F(R)F(S) \triangleq$ relator

(b) $(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \triangleq$ adj-all

(c) $S^\circ F(X)R \sqsubseteq X$



$S \cdot \neg S \setminus \triangleq$ adj-all

(a)

(a) $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq$ reduce, (6.2b), adj-all

§6.4. Fast exponentiation and modulus computation

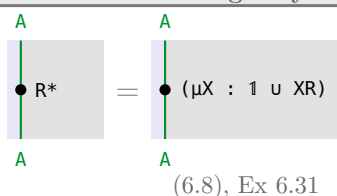
§6.5. Unique fixed points

§6.6. Sorting by selection

§6.7. Closure

$$R^* = (\mu X : 1 \cup XR) \quad (6.7a)$$

closing R by composing it on the left and closing it by composing on the right give the same relation R^*



(6.8), Ex 6.31

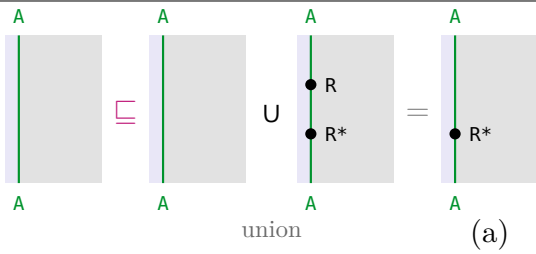
$$1 \in X \wedge XX \subseteq X \implies R \in X \iff R^* \in X$$

(6.7b)

 for a preorder X , X contains R exactly when it contains R^*

$$1 \in R^*$$

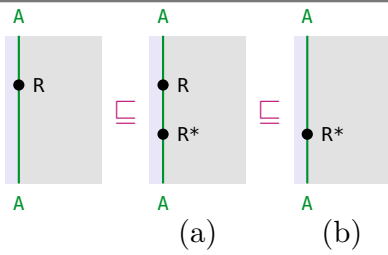
(6.7c)

 R^* is reflexive


(a) fixed point, Thm 6.1

$$R \in R^*$$

(6.7d)

 R^* contains R


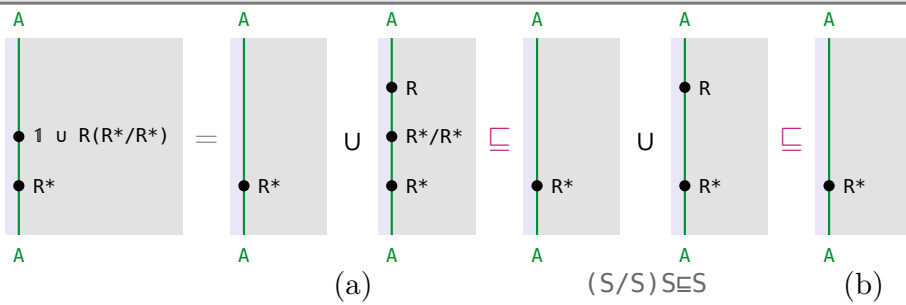
identity

 (a) $1 \in R^*$ — (6.7c)

(b) fixed point

$$R^* R^* \subseteq R^*$$

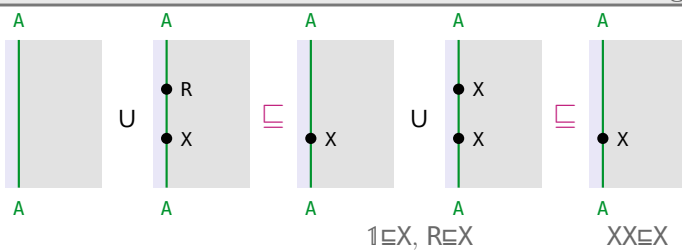
(6.7e)

 R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

 (a) composition distributes over $\cup$

 (b) $RR^* \subseteq R^*$ — (6.7d)

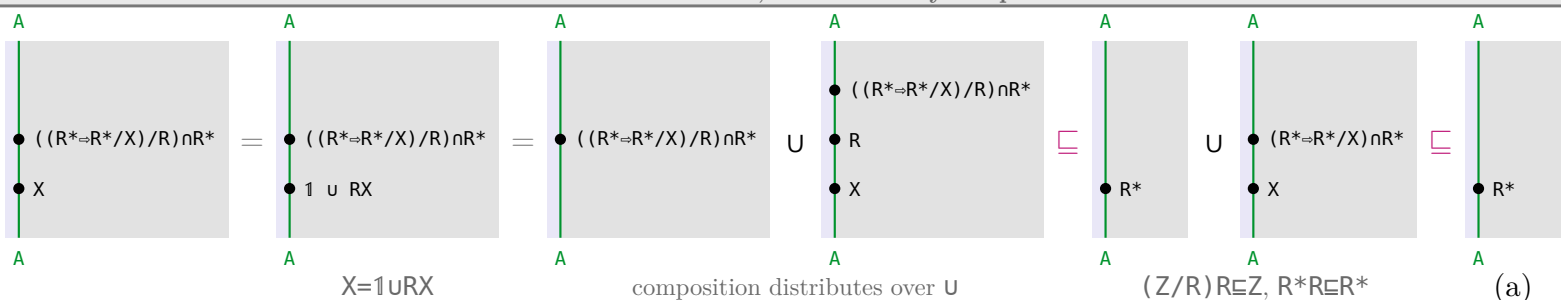
$$1 \in X \wedge XX \subseteq X \wedge R \in X \implies R^* \in X$$

(6.7f)

 a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

 $1 \in X, R \in X$
 $XX \subseteq X$

$$\text{inductive}(R) \wedge X = 1 \cup RX \implies X = R^*$$

(6.7g)

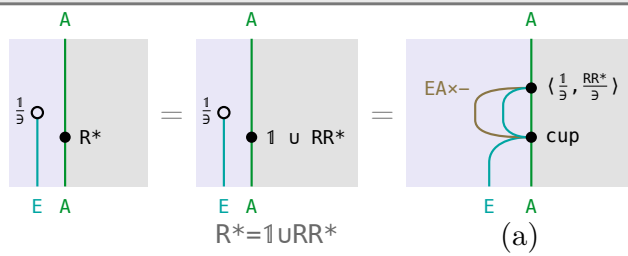
 when R admits induction, R^* is the only X equal to $1 \cup RX$

 (a) $(R^* = R^*/X) \cap R^* \subseteq R^*/X$
 $X = 1 \cup RX$

 composition distributes over $\cup$
 $(Z/R)R \subseteq Z, R^*R \subseteq R^*$

(a)

$$\frac{R^*}{S} = \left(\frac{1}{S}, \frac{RR^*}{S} \right) \text{cup} \quad (6.7h)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors



$$(a) \wedge (RuS) = \langle \wedge(R), \wedge(S) \rangle \text{cup}$$

$$R - \emptyset = R \quad (6.7i)$$

taking nothing away leaves R

$$R \cup (S - R) = R \cup S$$

beside R , only the part of S outside R adds anything

$$R - (S \cup T) = R - S - T$$

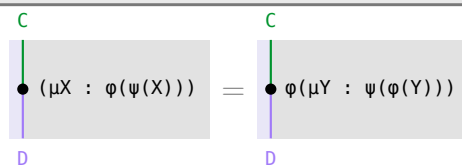
taking away S and then T takes away $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking T away from a union takes it away from each part

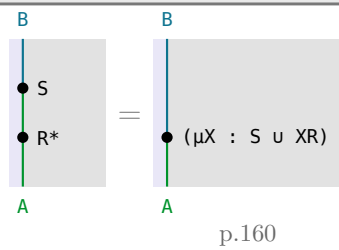
$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y))) \quad (6.7j)$$

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ



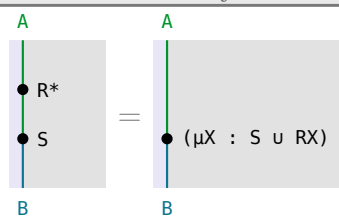
$$SR^* = (\mu X : S \cup XR) \quad (6.7k)$$

S followed by any number of R steps is the least X containing S and closed under a further R step



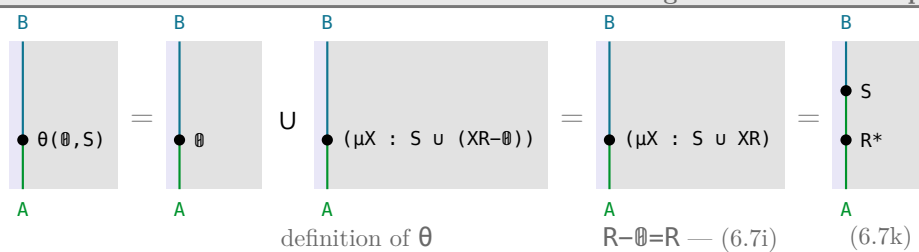
$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front



$$\theta(\emptyset, S) = SR^* \quad (6.7l)$$

started with nothing found and S to explore, θ computes SR^*



(6.7m)

$\theta(P, \emptyset) = P$
 with nothing left to explore, θ returns what it has found

definition of θ

(a)

$P \cup \emptyset = P$

(a) $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$

(6.7n)

$\theta(P, Q) = \theta(P \cup Q, QR - P - Q)$
 one round moves Q into the found part and explores the new R -successors of Q that are in neither

definition of θ

$QU S = QU(S - Q) \text{ --- (6.7i)}$

rolling --- (6.7j)

subtraction --- (6.7i)

definition of θ